Lecture 1

What is Bioinformatics?

And why epistemic awareness matters as much as technical skill

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Computing for Data-Driven Biology · 5BI00A
Master's Programme in Bioinformatics · Umeå University

Session Aims & ILO Links

1. Describe the origins, scope, and interdisciplinary nature of bioinformatics — ILO 1, 5
2. Explain what bioinformatics analyses can and cannot tell us — epistemic grounding — ILO 3, 13
3. Articulate the potential and specific risks of AI-assisted coding in research — ILO 13
4. Identify all programme ILOs and connect them to career trajectories — ILO 1–13

Part 1

Origins: How did bioinformatics come to exist?

Three Transitions That Created Modern Bioinformatics

1990–2003

The Human Genome Project

Demonstrated genome-scale sequencing was possible — and that interpreting the data required entirely new computational approaches. Established public data sharing as a scientific norm.

~2005–present

The NGS Revolution

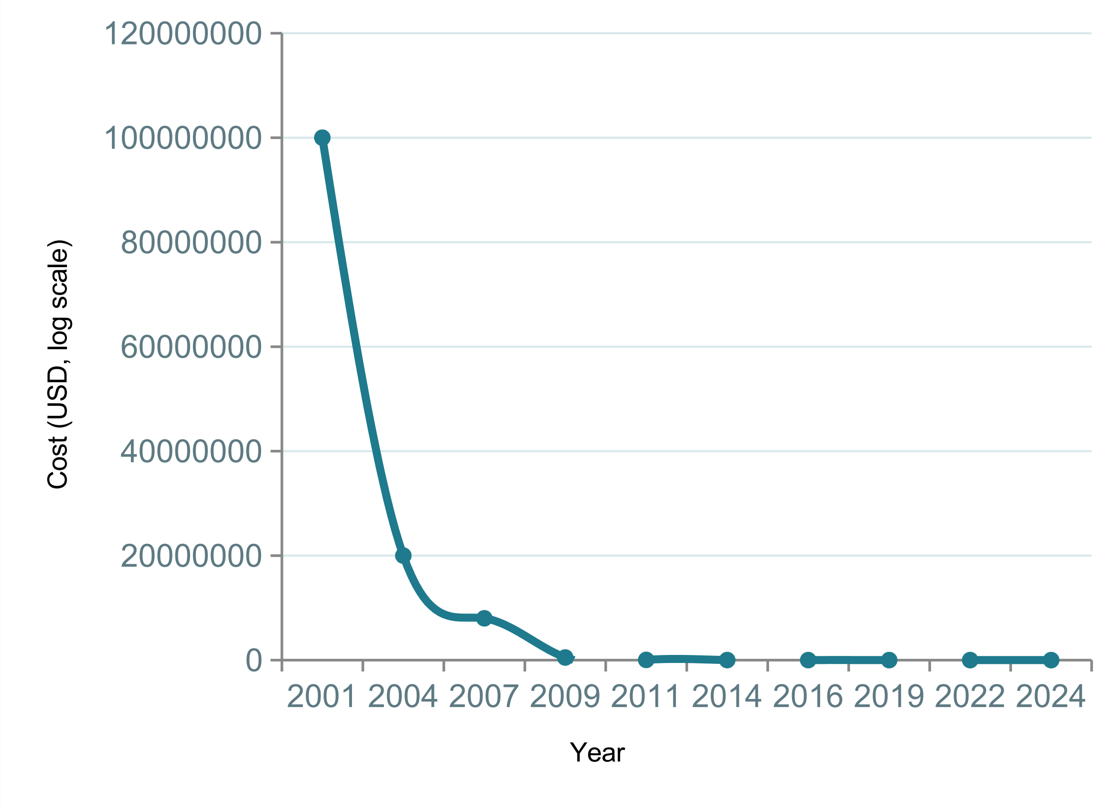
Sequencing costs dropped faster than Moore's Law. A $100M genome in 2001 costs ~$200 today. The bottleneck in biology shifted from generating data to making sense of it.

Now

The Multi-Omics Era

Simultaneous profiling of genome, transcriptome, proteome, epigenome, and metabolome. Extraordinary data complexity that no single discipline can interpret alone.

The Sequencing Cost Curve: Faster Than Moore's Law



$100M

Cost to sequence
one genome in 2001

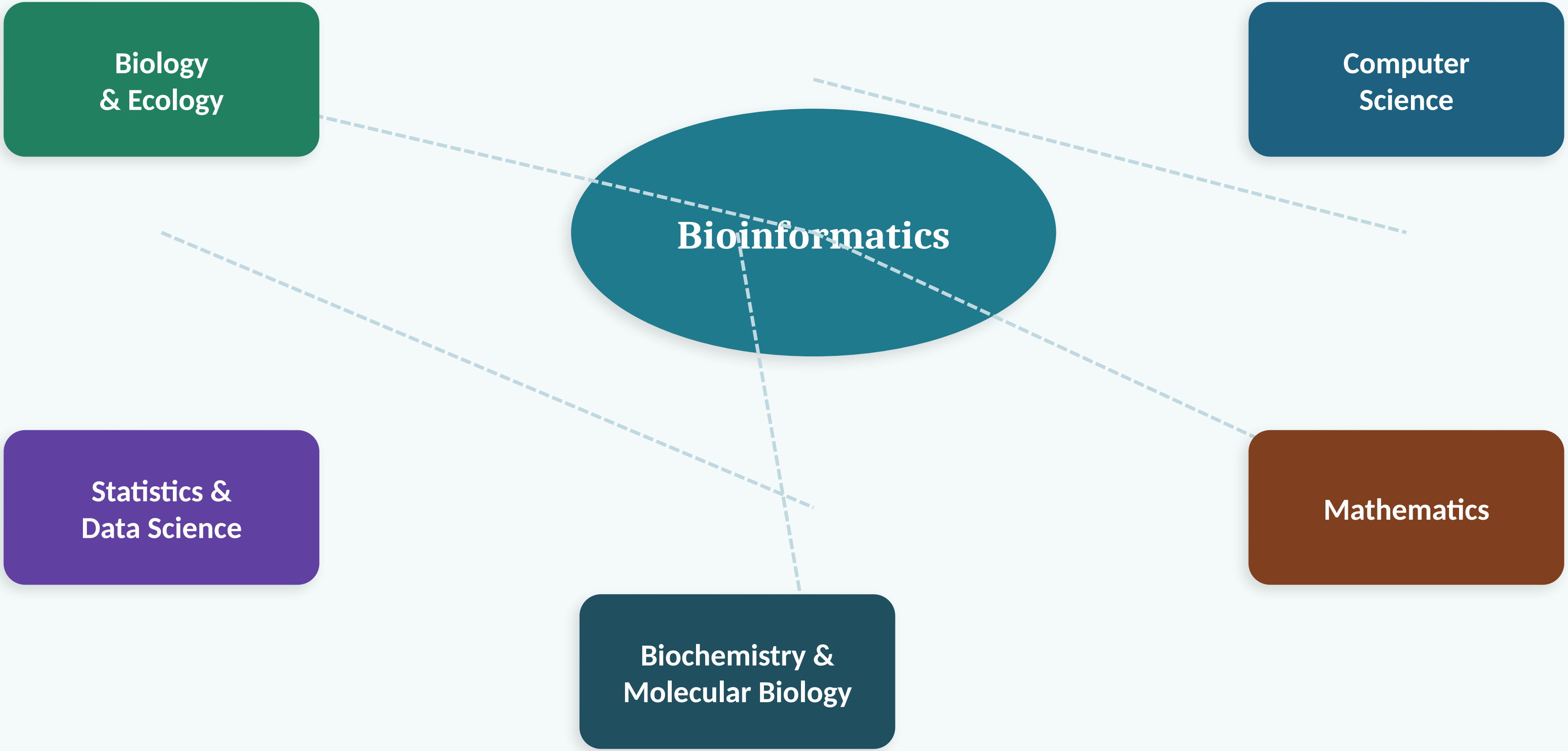
$200

Approximate cost
per genome today

>6 orders of magnitude

Cost reduction
2001 → 2024

Bioinformatics: An Interdisciplinary Field



No single practitioner is expert in all areas. Bioinformatics is practised at the interfaces.

Where Bioinformatics is Applied

🧬 Human Genomics & Medicine

Variant calling; cancer genomics; pharmacogenomics

🏥 Clinical Genetics

Diagnostic sequencing; variant interpretation

🌱 Plant Science & Agriculture

Crop improvement; climate adaptation; wood formation

🦠 Microbiology

Pathogen surveillance; microbiome; antimicrobial resistance

🌍 Ecology & Biodiversity

eDNA metabarcoding; SILVA; species monitoring

💊 Drug Discovery & Biotech

Target identification; protein structure; biomarkers

This programme spans all of these domains — the computational skills are transferable across all of them.

Part 2

What can — and cannot — bioinformatics tell us?

“

Bioinformatics analysis generates hypotheses that should be treated as such until supported by independent evidence.

This is not scepticism about the field — it is the mark of a rigorous scientist.

Four Reasons to Stay Epistemically Alert

01

Reference bias

Sequences, variants, and features that differ from the reference genome may be missed, misassembled, or incorrectly annotated.

02

Annotation circularity

Functional annotations are propagated computationally between databases. Errors compound and re-enter the literature.

03

Pipeline opacity

Each tool in a workflow makes invisible decisions. If you cannot explain every parameter, you do not fully understand your own results.

04

Absence ≠ non-existence

A non-significant result is not evidence of no biology. Absence of detection reflects the limits of the experiment, not of the organism.

Reference Bias: What the Reference Cannot See

The concept

Almost all genomic analyses compare reads to a **reference genome** — a single representative sequence derived from one (or few) individuals.

Anything substantially different from the reference may be:

- missed
- misassembled
- incorrectly annotated

Where it matters most

Population genomics

Reference genomes are often derived from historically well-resourced research communities; diversity may be systematically under-detected.

Non-model organisms

Reference genomes may be fragmented, incomplete, or absent entirely.

Structural variants

Large genomic rearrangements are poorly detected by short reads aligned to a linear reference.

Pan-genomes

Modern solution: graph-based pan-genome reference models capture population-level variation.

Annotation Circularity: When Errors Compound

Gene X characterised experimentally in species A → Annotation transferred by sequence homology → B → B's annotation re-used to annotate species C, D… → **Error in A's annotation now in dozens of genomes**

⚠ **Practical rule:**

Always check the evidence code for a database annotation.
Is it based on experimental evidence (EXP), computational prediction (IEA), or transferred annotation (ISS)?
These are not equivalent — and the distinction matters for how much you trust the result.

Part 3

Artificial intelligence: power, risk, and responsibility

AI Coding Tools: Genuinely Transformative

We actively encourage you to become fluent with AI-assisted coding.
These tools are already part of professional bioinformatics.

⚡ Accelerate routine tasks

Boilerplate scripts, pipeline scaffolding, format conversions

🔍 Interpret unfamiliar code

Understand what an inherited pipeline actually does

📝 Draft documentation

README files, inline comments, method descriptions

🐛 Debug error messages

Rapid triage of cryptic stack traces and tool outputs

🔗 Explore APIs and tools

Discover parameters, examples, and idiomatic usage

📚 Synthesise literature

Rapid orientation to unfamiliar domains and methods

The AI Risk Hierarchy in Bioinformatics

Level 1 — Inconvenience

Code fails immediately. You notice. Minor time lost.

Level 2 — Hidden error

Code runs but implements wrong logic. Results are wrong. You may not discover this until peer review or failed reproduction.

Level 3 — Compounding

Wrong results inform downstream analyses, grant applications, or clinical decisions. The error propagates through your work.

Level 4 — Systemic

AI-generated scientific content (hallucinated citations, fabricated results) enters the published literature. Already occurring.

Levels 1–2 are common. Levels 3–4 are already happening.

The core problem

AI doesn't make mistakes you can't detect. It makes mistakes you can't detect yet.

🏎️ **Formula 1 analogy**

An AI coding assistant hands you the keys to an F1 car.

You can go very fast.

Almost certainly in the wrong direction.

Possibly through a wall.

("AI COPILOT: No issues detected!")

An experienced bioinformatician reviewing AI-generated code immediately notices when a statistical test is misapplied, or coordinates are confused. A student without that foundation cannot yet perform that evaluation.

Learn to Crawl, Walk, and Run

🐌 CRAWL

Conceptual foundations

What does this analysis actually do?
What are its assumptions?
What could go wrong?

Where you are now. This is the most important stage.

🚶 WALK

Independent implementation

Build analyses from components you understand. Explain every decision. Verify outputs against known results.

Goal for Year 1 of this programme.

🚀 RUN

AI-accelerated expertise

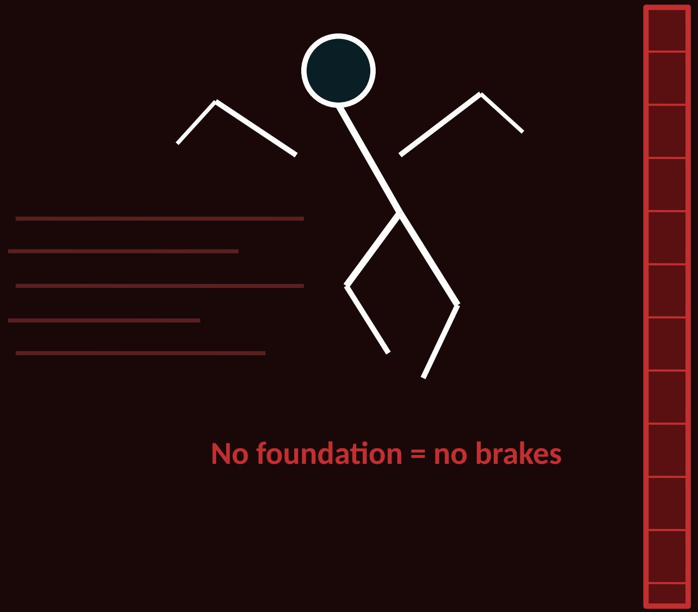
Use AI tools to accelerate work you already understand deeply enough to verify, evaluate, and take responsibility for.

Where AI tools genuinely transform productivity.

What happens when you skip straight to run

CRAWL | WALK | RUN

Skipped → jump to AI tools



No foundation = no brakes

Consequences of skipping

- ✗ Results look correct — you cannot tell they are wrong
- ✗ Real biological signals are missed entirely
- ✗ Errors compound through downstream analyses
- ✗ Work cannot be reproduced or defended
- ✗ In clinical settings: direct patient risk
- ✗ "The AI said it was fine" is not a defence

Domain knowledge is not a barrier to AI tools — it is what makes AI tools usable responsibly.

A concrete example — genomics without training

Intentional oversimplification — constructed to illustrate a conceptual point. Real clinical genomics has multiple layers of validation.

A member of the public has unexplained neurological symptoms. They sequence their genome via a consumer service and ask an AI tool to interpret the results.

AI-generated report

Variant: `rs00143267 (APXB4)`

Status: ⚠ **Likely pathogenic**

Condition: Progressive neuromotor syndrome

Supporting literature:

Smith J. et al. (2019). Nature Genetics 51: 234–241.

Johnson K. et al. (2021). N Engl J Med 385: 1123–1130.

Recommended: Cortramide 50 mg daily

Order from certified provider →

✓ Confidence: 94%

What domain knowledge reveals

✗ **rs00143267 is not pathogenic**

Frequency 31% in gnomAD — too common to cause a rare disease

✗ **Smith et al. (2019) — does not exist**

Not in PubMed or any database. Hallucinated citation.

✗ **Johnson et al. (2021) — wrong variant**

Paper exists but covers a different gene entirely.

✗ **Cortramide — dangerous here**

No evidence for this condition; serious drug interactions.

None of this was detectable from the report.
The output looked identical to a legitimate clinical document.

Practical Guidelines for AI Use in This Programme

✅ You are responsible for every line of code you submit, regardless of how it was generated. "The AI wrote it" is not an acceptable explanation for an error.

💬 Always be able to explain what your code does before submitting it. If you cannot explain it, you do not yet understand it sufficiently.

🧪 Verify outputs against expected results. Run analyses on test data with known answers before applying to real data.

📋 Document your use of AI tools. Transparency about AI assistance is an emerging norm in science — practise it from the beginning.

⚠️ AI tools hallucinate. They generate citations that do not exist, describe tools never implemented, and produce plausible-sounding but incorrect biological statements. Verify independently.

Part 4

The programme structure and your career trajectory

Year 1: Building the Foundations

Course	ECTS	Level
Computing for Data-Driven Biology ← You are here	7.5	G
Statistics and Data Science for Biology	7.5	G
Programming in Python for Bioinformatics	7.5	G
Statistics and Machine Learning in Bioinformatics	7.5	G
Quantitative Biology and Integrative Omics	7.5	A
Comparative Genomics	7.5	A
Functional Genomics Theory	7.5	A
Applied Functional Genomics	7.5	A

A = Advanced level   G = Basic level

Where Can This Programme Take You?

Service Facility Bioinformatician

e.g. Clinical Genomics Sweden

Key skills: Reproducible pipelines · FAIR data · Variant interpretation

Research Bioinformatician

Core facility or research group

Key skills: Statistical rigour · Domain knowledge · Scientific communication

Bioinformatics Scientist

Biotech / pharmaceutical industry

Key skills: Scalable pipelines · Software engineering · Data science

PhD Student & Future Academic

University research group

Key skills: Critical analysis · Methods development · Original contribution

Environmental Bioinformatician

Consultancy / governmental agency

Key skills: Metagenomics · Biodiversity informatics · eDNA analysis

Discussion: Five Questions

Take 2 minutes individually, then discuss in pairs.

1. A paper reports Gene X is "conserved across vertebrates" based on BLAST homology. What three things would you want to know before accepting this?

2. A collaborator sends you an AI-generated Python script. It runs without errors. What do you do before using the results in a manuscript?

3. Which is better supported by a bioinformatics analysis: (a) "Gene X is expressed in liver", or (b) "Transcripts mapping to Gene X are detected with mean coverage 47×"? Why?

4. Without looking anything up — what do you think Findable, Accessible, Interoperable, and Reusable mean in practice for a bioinformatics dataset?

5. What did you most recently read about AI in science? Was the framing balanced? What was missing?

What You Should Know After This Session

- ✓ Describe the historical context of bioinformatics: HGP, NGS, multi-omics
- ✓ Define bioinformatics as an interdisciplinary field and name contributing disciplines
- ✓ Give ≥ 2 examples of application domains, spanning human health and environmental sciences
- ✓ Explain ≥ 2 specific ways bioinformatics analyses can mislead (reference bias, annotation circularity, etc.)
- ✓ Articulate the epistemic principle: bioinformatics generates hypotheses, not definitive facts
- ✓ Describe the AI risk hierarchy and why Level 2 errors are particularly dangerous for beginners
- ✓ Identify all course ILOs and link them to at least one career trajectory

Next: FAIR and Open Science (Lecture 14) — the principles that make bioinformatics reproducible and trustworthy.